**CARD 1 OF 28** CLASSES **QUESTION**

Spec says “Class III standpipe.” **What hose outlets does that put on your plans**, and who is each one for?

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**CARD 2 OF 28** TYPES **QUESTION**

A standpipe is **full of water** (fed off a small domestic tap to keep it wet), but the **only** way to get the design flow is the pumper at the FDC. **What type is it?**

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**CARD 3 OF 28** HIGH-RISE **QUESTION**

Top occupiable floor is **82 ft** above the lowest fire-truck access level. Owner wants a **manual dry** Class I standpipe to save the pump. **Allowed?**

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**CARD 4 OF 28** WET / DRY **QUESTION**

A heated 4-story (non-high-rise) office gets a **Class III** system. **Can any of it be manual or dry?**

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*FSM · NICET L3 — NFPA 14 (2019) Standpipes · sheet 1 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

**CARD 2 OF 28** TYPES **ANSWER**

**Manual wet.** Water in it all the time, but the FDC is the only source of system demand.
**Manual dry** = no attached supply at all, FDC only.
**Semiautomatic dry** = attached supply behind a device like a deluge valve, opened by a remote control device at the hose connection.

*NFPA 14 (2019) · Ch. 3*

**Trap:** "it has water, so it's automatic wet." Automatic means the attached supply can deliver the full system demand on its own.



**CARD 1 OF 28** CLASSES **ANSWER**

**Class I** — 2½ in. hose connections for the fire department (heavy streams).
**Class II** — 1½ in. hose stations for trained personnel (or a connection for the FD on initial response).
**Class III** — **both**: 1½ in. stations *and* 2½ in. connections.

*NFPA 14 (2019) · Ch. 5*

**Trap:** treating Class III as "a bigger Class I." It is Class I + Class II on the same system.



**CARD 4 OF 28** WET / DRY **ANSWER**

**Class I:** wet unless the piping is exposed to freezing.
**Class II / III with 1½ in. stations:** automatic wet — dry allowed only where freezing *and* a trained fire brigade runs it without the FD.
**Non-high-rise Class III:** the Class I part may be **manual**; the Class II part must stay **automatic**.

*NFPA 14 (2019) · Ch. 5*

**Trap:** making the whole Class III manual. Only the 2½ in. (Class I) side gets that break.



**CARD 3 OF 28** HIGH-RISE **ANSWER**

**No.** Over 75 ft to an occupiable floor makes it a high-rise, and every required Class I standpipe in a high-rise must be **automatic or semiautomatic** — including partial-height and horizontal standpipes.
**One exception:** an **open parking garage** may stay manual if its top floor is no more than **150 ft** above the lowest FD vehicle access.

*NFPA 14 (2019) · Ch. 3 · Ch. 5*

**Trap:** measuring to the roof or from grade. It is the **floor of the occupiable story**, measured from the lowest level of FD vehicle access.



*sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back*

**CARD 5 OF 28** HOSE CONN. LOCATION **QUESTION**

Laying out Class I outlets in a building with **two required exit stairs** and one convenience stair between floors 2 and 3. **Where do the outlets go?**

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**CARD 6 OF 28** HORIZONTAL EXITS **QUESTION**

A **sprinklered** floor is split by a horizontal exit. All floor area is within **185 ft** of an exit-stair hose connection on the same side of the horizontal exit. **Do you need both horizontal-exit outlets?**

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**CARD 7 OF 28** ROOF **QUESTION**

**None** of the stairs reach the roof, and the roof is **flat**. Is a **roof** hose connection required? What if the roof were a 6-in-12 pitch?

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**CARD 8 OF 28** TRAVEL DISTANCE **QUESTION**

Big-box floor: the far corner is **230 ft** from the nearest stair hose connection. **Is an extra Class I outlet needed** — sprinklered vs. unsprinklered?

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**CARD 6 OF 28** HORIZONTAL EXITS **ANSWER**

**No — one can be dropped.** The default is a hose connection on **each side** of the wall at the horizontal exit opening. The one on the **other side** may be omitted when all floor areas reach a stair connection on the same side within **200 ft sprinklered** / **130 ft nonsprinklered**. 185 < 200.

*NFPA 14 (2019) · Ch. 7*

**Trap:** same building unsprinklered → 185 > 130, the outlet stays.



**CARD 5 OF 28** HOSE CONN. LOCATION **ANSWER**

A hose connection at **each main floor landing** of every **required** exit stair, with a **separate standpipe in each required exit stair**.
Intermediate (half-) landings only where the AHJ / local FD asks for them.
A **nonrequired** stair linking two adjacent floors needs **none**.

*NFPA 14 (2019) · Ch. 7*

**Trap:** defaulting to the intermediate landing. Main floor landing is the rule; intermediate is the FD's option.



**CARD 8 OF 28** TRAVEL DISTANCE **ANSWER**

**Sprinklered:** limit **200 ft** → 230 > 200, **add one** (location approved by the AHJ, who may also waive it).
**Unsprinklered:** limit **150 ft** → **add one**.
Measured **from the hose connection**. **Open parking garage with manual dry:** both drop to **130 ft**. A roof not meant for occupancy doesn't count.

*NFPA 14 (2019) · Ch. 7*

**Trap:** mixing up 200/130 (horizontal-exit omission) with 200/150 (added outlets). Unsprinklered is 130 at a horizontal exit but 150 here.



**CARD 7 OF 28** ROOF **ANSWER**

**Flat roof: yes** — with no stair access, the outlet goes on the roof itself.
**6-in-12: no** — slope of **4 in 12 or steeper** drops it.
Where a stair *does* reach the roof, the outlet goes at that stair's **highest landing** instead.

*NFPA 14 (2019) · Ch. 7*

**Trap:** forgetting the roof outlet altogether because no stair goes up there. That is exactly the case that requires one.



*sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back*

**CARD 9 OF 28** CLASS II / III **QUESTION**

Class II hose stations: **how far apart?** And in a **fully sprinklered** Class III building, how do you get out of installing them?

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**CARD 10 OF 28** OUTLET HEIGHT **QUESTION**

Detailing the hose valve on the riser. **How high off the landing**, measured to what? Any clearance rules?

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**CARD 11 OF 28** RESIDUAL PRESSURE **QUESTION**

Hydraulically designed system: what **residual pressure** must you deliver at the most remote **2½ in.** outlet? The most remote **1½ in.**? And for a **manual** standpipe?

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**CARD 12 OF 28** CLASS I FLOW **QUESTION**

A 12-story **sprinklered** office (floors well under 80,000 $ft^2$) has **3 standpipes**. **Minimum Class I system flow?**

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**CARD 10 OF 28** OUTLET HEIGHT **ANSWER**

**3 ft minimum to 5 ft maximum** above the floor, measured to the **center of the hose valve**.
Not blocked by the stair door (open *or* closed) or anything else on the landing, and at least **3 in.** between the valve handle and any object through its full swing.

*NFPA 14 (2019) · Ch. 4 · Ch. 7*

**Trap:** measuring to the outlet cap or the bottom of the valve. Center of the valve.

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**CARD 9 OF 28** CLASS II / III **ANSWER**

**Class II:** every part of the floor within **130 ft** of a station with 1½ in. hose, or **120 ft** with smaller hose — measured **along the path of travel** from the connection.
**Class III, sprinklered throughout (NFPA 13/13R):** skip the Class II stations if every Class I outlet is **2½ in.** with a **2½ × 1½ in. reducer** and a **chained cap**. Then the Class II travel limit doesn't apply.

*NFPA 14 (2019) · Ch. 5 · Ch. 7*

**Trap:** measuring Class II distance as a straight line on plan. It is walking distance.

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**CARD 12 OF 28** CLASS I FLOW **ANSWER**

**1000 gpm.**
Most remote standpipe: **500 gpm** (through its two most remote 2½ in. outlets).
Each additional standpipe: **250 gpm** → 2 × 250 = 500.
500 + 500 = **1000 gpm** — which is also the sprinklered maximum, so a 4th standpipe adds nothing.

*NFPA 14 (2019) · Ch. 7*

**Trap:** 500 gpm per standpipe (= 1500). Only the first one is 500.

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**CARD 11 OF 28** RESIDUAL PRESSURE **ANSWER**

**2½ in.: 100 psi** residual at the most remote connection.
**1½ in.: 65 psi** at the outlet of the most remote connection.
**Manual:** 100 psi at the outlet of the most remote 2½ in. valve, with the calc **ending at the FDC** — and that FDC pressure can't exceed the working pressure of the system parts.

*NFPA 14 (2019) · Ch. 7*

**Trap:** using 65 psi for Class I. 65 is the 1½ in. (Class II) number; Class I is 100.

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**CARD 13 OF 28** CLASS I FLOW **QUESTION**

An **unsprinklered** building, floors under 80,000 ft², has **6 standpipes**. **What flow do you size the supply for?**

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**CARD 14 OF 28** HORIZONTAL STANDPIPE **QUESTION**

A **horizontal standpipe** runs across a podium level and feeds **4 hose connections** on that floor. **What does its flow become**, and how is it calculated?

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**CARD 15 OF 28** HYDRAULIC CALC **QUESTION**

Setting up the Class I calc. **Where do you put the flows**, and do the other standpipes have to be balanced to the high pressure at their tie-in?

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**CARD 16 OF 28** CLASS II FLOW **QUESTION**

A Class II system has **3 standpipes and 30 hose stations**. **Total flow?** And what is the most any single outlet is ever required to flow?

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**CARD 14 OF 28** HORIZONTAL STANDPIPE **ANSWER**

**750 gpm** minimum for the most demanding horizontal standpipe — triggered at **3 or more** hose connections on any floor.

Calc: **250 gpm at each of the three** most remote connections (plus 250 at the connection point of each other standpipe) at the Ch. 7 minimum residual pressure.

*NFPA 14 (2019) · Ch. 7*

**Trap:** using the vertical-riser rule (500 gpm, two outlets) on a horizontal run with three or more outlets.



**CARD 13 OF 28** CLASS I FLOW **ANSWER**

**1250 gpm.**

500 + (5 × 250) = 1750 gpm by the per-standpipe rule…

…but the total is capped at **1250 gpm** when not sprinklered throughout (**1000 gpm** when sprinklered throughout per NFPA 13).

*NFPA 14 (2019) · Ch. 7*

**Trap:** carrying 1750 into the pump selection. The cap governs.



**CARD 16 OF 28** CLASS II FLOW **ANSWER**

**100 gpm, total.** 100 gpm at the most remote station; **no added flow** for more stations, and common supply piping is sized for 100 gpm.

Per-outlet maximum: **250 gpm** from a 2½ in. connection, **100 gpm** from a 1½ in. connection.

*NFPA 14 (2019) · Ch. 7*

**Trap:** adding 100 per standpipe like the Class I 250 rule. Class II never stacks.



**CARD 15 OF 28** HYDRAULIC CALC **ANSWER**

**250 gpm at each of the two** most remote outlets on the remote standpipe, and **250 gpm at the connection point** of each other standpipe — all at the 100 psi minimum residual.

**No balancing** of the added standpipe flows to the higher junction pressure.

Risers ending at **different floors**: run a separate calc per level, counting only standpipes that exist on that level.

*NFPA 14 (2019) · Ch. 7*

**Trap:** balancing every tie-in like a sprinkler grid. The standard says you don't have to.

**CARD 17 OF 28** COMBINED SYSTEM **QUESTION**

**Fully sprinklered** building, combined system. Sprinkler demand with hose stream = **650 gpm**. Standpipe demand = **1000 gpm**. **What does the water supply have to deliver?**

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**CARD 18 OF 28** COMBINED SYSTEM **QUESTION**

**Partially** sprinklered building, combined system. The sprinklered area is **ordinary hazard** and calcs at **420 gpm**. **How much do you add to the standpipe flow?** What if it were **light hazard** calcing at 210 gpm?

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**CARD 19 OF 28** PIPE SIZES **QUESTION**

**Minimum pipe sizes**, Class I / III: a plain standpipe, a combined riser in a **partially** sprinklered building, a combined riser in a **fully** sprinklered hydraulically designed building, and a branch line?

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**CARD 20 OF 28** MAX PRESSURE **QUESTION**

Tall tower: at the low-zone base, pump churn plus static head comes to **430 psi**. **Is that allowed?** What about an express main to the upper zone?

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**CARD 18 OF 28** COMBINED SYSTEM **ANSWER**

Add the **lesser** of the calculated sprinkler demand or the fixed number:
**OH:** lesser of 420 or **500 gpm** → add **420 gpm**.
**LH:** lesser of 210 or **150 gpm** → add **150 gpm**.

*NFPA 14 (2019) · Ch. 7*

**Trap:** always adding the flat 150 / 500. It is whichever is **less**.



**CARD 17 OF 28** COMBINED SYSTEM **ANSWER**

**1000 gpm.** The supply is the **greater** of the sprinkler demand (hose stream included) or the standpipe demand — not the sum.
No separate sprinkler demand is required, and the sprinkler hose demand is **not added** to the standpipe calc.

*NFPA 14 (2019) · Ch. 7*

**Trap:** 650 + 1000 = 1650. That is the partially-sprinklered mindset (next card).



**CARD 20 OF 28** MAX PRESSURE **ANSWER**

**Not in a zone with outlets.** Max pressure anywhere in the system, at any time: **400 psi** — that is the reason for zoning.
**Express main** feeding higher zones may run above 400 psi per its material listings or AHJ approval — but **no hose valves** on any part above 400 psi.

*NFPA 14 (2019) · Ch. 7*

**Trap:** checking only the flowing (residual) pressure. "At any time" includes churn / static.



**CARD 19 OF 28** PIPE SIZES **ANSWER**

**Standpipe:** 4 in.
**Combined, partially sprinklered:** 6 in.
**Combined, fully sprinklered (NFPA 13/13R), hydraulically designed to the 100 psi rule:** 4 in.
**Branch line:** sized by hydraulics, never under 2½ in.

*NFPA 14 (2019) · Ch. 7*

**Trap:** "combined = 6 in., always." 6 in. is the partially-sprinklered case only.



*sheet 5 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back*

**CARD 21 OF 28** PRDS **QUESTION**

Floor 3 of a tower: **2½ in.** outlet sees **190 psi static**; a **1½ in.** station sees **120 psi residual**. **Which need a pressure-regulating device**, and set to what?

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**CARD 22 OF 28** PRDS **QUESTION**

One PRD serves **more than two** hose connections downstream. **What has to show up on your riser detail?**

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**CARD 23 OF 28** ZONES **QUESTION**

Two-zone tower. **Is there a maximum zone height in feet?** And what does each zone need for supply?

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**CARD 24 OF 28** FDC — NUMBER **QUESTION**

Single-zone **high-rise**, standpipe demand **1000 gpm**. **How many FDCs and how many 2½ in. inlets?**

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**CARD 22 OF 28** PRDS **ANSWER**

- PRDs **in series**, so one failure can't pass over **175 psi** downstream
- equal-size **bypass** with a **normally closed** valve
- PRD no higher than **7 ft 6 in.** above the floor
- **inlet and outlet gauges**, a relief valve, remote high-pressure supervision (NFPA 72)
- a **drain riser** beside each standpipe with PRDs: at least **3 in.** for 2½ in. PRDs, **2 in.** for 1½ in.

*NFPA 14 (2019) · Ch. 7*

**Trap:** one master PRD with no backup. A single failure would put full pump pressure on every outlet below it.



**CARD 21 OF 28** PRDS **ANSWER**

**Both.**
**2½ in.:** static over **175 psi** → listed PRD holding static *and* residual to 175 psi.
**1½ in.:** residual over **100 psi** → listed PRD holding residual to 100 psi at design flow (and a 1½ in. with static over 175 also gets the 175 limit).
The 1½ in. rule skips the outlet on a 2½ × 1½ reducer and a 1½ in. connection with no hose required.

*NFPA 14 (2019) · Ch. 7*

**Trap:** checking the 2½ in. on residual. Its trigger is **static** over 175.



**CARD 24 OF 28** FDC — NUMBER **ANSWER**

**Inlets:** one 2½ in. per 250 gpm → 1000 / 250 = **4 inlets** (or an approved large-diameter connection). They may be split across FDCs.
**FDCs: two per zone** in a high-rise (also any building over **900 ft** perimeter), on opposite corners or as far apart as possible. Each FDC: at least two 2½ in. inlets.
Any standpipe system: at least one.

*NFPA 14 (2019) · Ch. 4 · Ch. 7*

**Trap:** one Siamese because "that's what's on every job." High-rise = two per zone.



**CARD 23 OF 28** ZONES **ANSWER**

**No height in feet** in the 2019 text — zones exist to keep pressure within the **400 psi** limit / component ratings.
Each zone: its **own pump** or its own outlet of a multistage, multiport pump (series pumps OK; PRDs per the >2-outlet rules may replace lower-zone pumps).
Each zone above the low zone: **two or more separate, direct supply pipes**, each able to carry the full demand.
If FD pumpers can't reach a high zone through the FDC: **high-level storage + pumping** (or AHJ-accepted means).

*NFPA 14 (2019) · Ch. 7*

**Trap:** quoting a fixed zone height. 2019 governs by pressure, not feet.

**CARD 25 OF 28** FDC — LOCATION **QUESTION**

Placing the FDC on the site plan. **Distance limits, mounting height, and what the sign says** — inlet pressure required is 165 psi.

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**CARD 26 OF 28** DRY STANDPIPES **QUESTION**

**Automatic dry** standpipe in an unheated parking structure works out to **900 gal** of pipe on one dry pipe valve. **Legal?**

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**CARD 27 OF 28** WATER SUPPLY **QUESTION**

Automatic Class I system: **how long** must the supply hold the demand, and **who** has to be able to supply that demand?

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**CARD 28 OF 28** VALVES **QUESTION**

Valve checklist for the plans: **riser isolation, a 55 ft branch line, a combined-system sprinkler tap, the FDC line.** What goes where?

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**CARD 26 OF 28** DRY STANDPIPES **ANSWER**

**Only if it proves water time.** One dry pipe valve controls **750 gal** max — unless water reaches the **most remote hose connection in 3 minutes or less**, from normal air pressure with that connection fully open. A listed quick-opening device may be used to get there.
Also: dry standpipes get **one level** of interconnection only, and are supervised on air pressure (NFPA 72).

*NFPA 14 (2019) · Ch. 5 · Ch. 6 · Ch. 7*

**Trap:** treating 750 gal as a hard cap. Water delivery time is the way past it.



**CARD 25 OF 28** FDC — LOCATION **ANSWER**

**Within 50 ft** of the street or nearest FD apparatus access, visible and recognizable.
**Within 100 ft** of a hydrant on an approved supply (AHJ may allow more).
**18 in. to 48 in.** above grade.
Sign "STANDPIPE," letters **≥1 in.** (add AUTOSPKR if it feeds sprinklers; manual systems say manual + wet/dry). **165 psi > 150** → a required-pressure sign is needed.

*NFPA 14 (2019) · Ch. 6 · Ch. 7*

**Trap:** skipping the pressure sign. It can be left off only at **150 psi or less**.



**CARD 28 OF 28** VALVES **ANSWER**

**Every standpipe** (manual dry and horizontal included): an isolation valve so one can be shut without killing the others.
**Branch line:** 55 ft > **40 ft** (along the pipe) to the remote outlet → listed indicating valve at the standpipe.
**Sprinkler tap off a combined riser:** its own control valve *and* check valve.
**FDC line: no isolation valve** between the FDC and the system piping; listed check valve.
Supply / isolation valves supervised open.

*NFPA 14 (2019) · Ch. 6*

**Trap:** putting a shutoff on the FDC line "for service." Not permitted.



**CARD 27 OF 28** WATER SUPPLY **ANSWER**

**30 minutes** minimum at the Ch. 7 flow and pressure.
The demand must be suppliable **through each FDC**; and where an automatic / semiautomatic supply is required, **independently** by the attached supply **and** by each FDC.

*NFPA 14 (2019) · Ch. 7 · Ch. 9*

**Trap:** sizing the FDC piping only to "help" the pump. Each FDC alone must carry the full demand.



*sheet 7 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back*